

Financial Time Series
Generation

Financial Time Series Ex-
trapolation

Statistical Frameworks

ARIMA, GARCH

Deep Learning Models

Hybrid LSTMs: LSTM, RNN, PEC-W, CEEMDAN-Informer-LSTM, SGP-LSTM, IRM

Generative Learning Models

GAN-based Models: GAN, WaveGAN, N-BEATS-GAN, WGAN-GP

Diffusion Models: DDPM, TimeGrad, TimeDiT

Time-series Foundation Models

Fine-tuned LLM: Chronos, OneFitsAll, TimeHF

Edited LLM: StockTime, LLM4FTS, LLM-PS, Time-LLM, CALF

Emerging FMs: FinCast

Financial Time Series Im-
putation

Statistical Methods

Temporal Disaggregation & Interpolation:
Chow-Lin, Kernel Ridge Regression (RKHS), KNN, MICE, Random Forest

Machine Learning

Factorization Methods: NMTucker, RegTensor

Deep Learning Models

Sequence Reconstruction Models: BiGRU-BiLSTM with GMDH Decoder

Generative Learning Models

Adversarial Imputation Models:
GAIN, CWGAN-GP, ImputeGAN, AttnWGAIN

Diffusion-based Imputation Models:
CSDI, Score-CDM, SSSD, TimeDiT, LSCD

Advanced Frameworks

Foundation Models & Dual Graph: LLM-Forest, MTabGen, Dual-Bipartite Graph Network

Financial Time Series Syn-
thesis

Micro-level Synthesis

Classical Models: SDEs (GBM, Heston Model)

Generative Adversarial Networks (GANs):
WGAN, WGAN-GP, QuantGAN, RSQGAN, DAT-CGAN, Jinkou, TAIL-GAN

Diffusion / SDE-based Models: TimeGrad, SGM, CoFinDiff, FinDiff

Macro-level Simulation*

Conditional GANs:
CoMeTS-GAN, CTS-GAN, Market-GAN, MacroSynth, GAN-MoEx

Conditional DMs: TRADES, DiGA

Large Market Models: MarS